

Advanced (Carolina Reaper)

- Q1.** What is the x -coordinate of the vertex of the parabola $y = x^2 + 4x + 4$?
- (A) -2
(B) -1
(C) 0
(D) 1
(E) 2
- Q2.** The parabola $y = ax^2 + bx + c$ has an axis of symmetry given by the equation $x =$
- (A) $-b/a$
(B) $-b/2a$
(C) $b/2a$
(D) $-c/b$
(E) $-c/2b$
- Q3.** What are the x -intercepts of the parabola $y = x^2 - 4$?
- (A) $(-2, 0)$ and $(2, 0)$
(B) $(-1, 0)$ and $(1, 0)$
(C) $(-3, 0)$ and $(3, 0)$
(D) $(-4, 0)$ and $(4, 0)$
(E) $(-5, 0)$ and $(5, 0)$
- Q4.** The parabola $y = x^2 - 2x - 3$ opens
- (A) upward
(B) downward
(C) to the left
(D) to the right
(E) not at all
- Q5.** The vertex of the parabola $y = ax^2 + bx + c$ is (h, k) . What is the value of a if the parabola opens downward and has a maximum value of 5?
- (A) -1
(B) -2
(C) -3
(D) -4
(E) -5
- Q6.** What is the equation of the axis of symmetry of the parabola $y = x^2 + 2x + 1$?
- (A) $x = -1$
(B) $x = -2$
(C) $x = -3$
(D) $x = 0$
(E) $x = 1$
- Q7.** The parabola $y = x^2 - 4x + 3$ has a minimum value of
- (A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q8.** What are the y -intercepts of the parabola $y = x^2 + 1$?
- (A) $(-1, 0)$ and $(1, 0)$
(B) $(-2, 0)$ and $(2, 0)$
(C) $(-3, 0)$ and $(3, 0)$
(D) $(-1, 1)$ and $(1, 1)$
(E) $(-2, 2)$ and $(2, 2)$

Open Ended Questions (FRQ)

- Q9.** Graph the parabola $y = x^2 - 2x - 3$ and identify its vertex, axis of symmetry, x -intercepts, and y -intercepts.
- Q10.** Find the equation of the parabola that has a vertex at $(2, 3)$, an axis of symmetry given by the equation $x = 2$, and a y -intercept at $(0, 1)$. Show your work.

Chef's Tips

- Make sure to provide students with graph paper.
- Encourage students to use a ruler or straightedge to draw their graphs.
- Remind students to label all important features on their graphs.